

Mutation & Pedigree Analysis

Worksheet · PU-II Biology, Chapter 4

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Worksheet – Mutation & Pedigree Analysis

15 questions · answer key on the last page

Q1. Mutation is best defined as:

- (a) A change in phenotype without any change in DNA
- (b) An alteration of DNA sequence, resulting in changes in genotype and phenotype
- (c) The blending of two parental phenotypes
- (d) The independent assortment of chromosomes

Q2. Which of the following is a physical mutagen?

- (a) Nitrous acid
- (b) UV radiation
- (c) Helicobacter (bacteria)
- (d) HPV (virus)

Q3. Which of the following is a chemical mutagen mentioned in the source material?

- (a) Gamma rays
- (b) Nitrous acid
- (c) UV radiation
- (d) HPV

Q4. A mutation resulting from a change in a single base pair of DNA is called a:

- (a) Frameshift mutation
- (b) Point mutation
- (c) Chromosomal aberration
- (d) Polygenic mutation

Q5. Sickle cell anaemia is an example of which type of mutation?

- (a) Frameshift mutation
- (b) Point mutation
- (c) Aneuploidy
- (d) Deletion mutation

Q6. A frameshift mutation results from:

- (a) A substitution of a single base pair only
- (b) Deletion or insertion of a base pair in the DNA sequence
- (c) Duplication of an entire chromosome
- (d) Fusion of two chromosomes

Q7. Pedigree analysis is used to:

- (a) Determine the number of chromosomes in a cell
- (b) Analyse the inheritance of a trait across generations of a family, represented as a family tree
- (c) Measure the recombination frequency between two genes
- (d) Identify the blood group of an individual directly

Q8. In pedigree charts, a square symbol represents:

- (a) A female
- (b) A male
- (c) An individual of unspecified sex
- (d) A deceased individual

Q9. In pedigree charts, a circle symbol represents:

- (a) A male
- (b) A female

- (c) A carrier
- (d) An affected individual

Q10. In pedigree charts, a diamond symbol represents:

- (a) A male
- (b) A female
- (c) An individual of unspecified sex
- (d) A miscarriage

Q11. In pedigree charts, a completely filled (shaded) symbol represents:

- (a) A carrier individual
- (b) An affected individual
- (c) A deceased individual
- (d) Identical twins

Q12. A double horizontal line joining a square and a circle in a pedigree indicates:

- (a) A normal mating
- (b) Mating between relatives (consanguineous mating)
- (c) Divorce
- (d) Dizygotic twins

Q13. A diagonal line drawn through a pedigree symbol indicates that the individual is:

- (a) Affected
- (b) A carrier
- (c) Deceased
- (d) The propositus

Q14. An arrow pointing to an individual in a pedigree chart indicates:

- (a) A deceased person
- (b) The propositus/proposita, i.e. the starting point of the genetic study
- (c) A carrier
- (d) Consanguineous mating

Q15. A number written inside a diamond symbol in a pedigree chart typically represents:

- (a) The generation number
- (b) That many unaffected offspring of unspecified sex
- (c) The age of the individual
- (d) The chromosome number

Answer Key & Explanations

- Q1.** (b) An alteration of DNA sequence, resulting in changes in genotype and phenotype — A mutation is an alteration of the DNA sequence itself, which can lead to changes in both genotype and phenotype.
- Q2.** (b) UV radiation — UV radiation, along with ionizing radiation and gamma rays, is a physical mutagen; nitrous acid is a chemical mutagen and HPV/Helicobacter are biological mutagens.
- Q3.** (b) Nitrous acid — Nitrous acid is a chemical mutagen that acts as a deaminating agent, changing cytosine to uracil in DNA.
- Q4.** (b) Point mutation — A point mutation arises from a change in a single base pair of DNA, as seen in sickle-cell anaemia.
- Q5.** (b) Point mutation — Sickle cell anaemia results from a point mutation, a single base-pair change in the beta globin gene.
- Q6.** (b) Deletion or insertion of a base pair in the DNA sequence — Frameshift mutations arise from the deletion or insertion of a base pair, which shifts the reading frame of the DNA sequence.
- Q7.** (b) Analyse the inheritance of a trait across generations of a family, represented as a family tree — Pedigree analysis traces the inheritance of a trait, abnormality, or disease across several generations of a family, represented as a family tree.
- Q8.** (b) A male — In standard pedigree notation, a square represents a male individual, while a circle represents a female.
- Q9.** (b) A female — A circle represents a female individual in standard pedigree chart notation.
- Q10.** (c) An individual of unspecified sex — A diamond symbol is used in pedigree charts to represent an individual whose sex is not specified.
- Q11.** (b) An affected individual — A fully filled or shaded symbol in a pedigree chart indicates that the individual is affected by the trait being studied.
- Q12.** (b) Mating between relatives (consanguineous mating) — A double horizontal line connecting a square and a circle represents consanguineous mating, i.e. mating between relatives.
- Q13.** (c) Deceased — A diagonal line through a symbol in a pedigree chart indicates that the individual is deceased.
- Q14.** (b) The propositus/proposita, i.e. the starting point of the genetic study — An arrow points to the propositus (male) or proposita (female), the individual who is the starting point of the genetic/pedigree study.
- Q15.** (b) That many unaffected offspring of unspecified sex — A number inside a diamond in a pedigree chart indicates that many unaffected offspring of unspecified sex.